CHAPTER 2

Basic Structures: Sets, Functions, Sequences, and Sums

2.1 Sets

2.2 Set Operations

2.3 Functions

2.4 Sequence and Summations

2.5 Cardinality of Sets

[The slides are courtesy of Prof. Lanfen Lin]



Some Concepts of Sets

【Definition】A *set* is an unordered collection of objects.

【Definition】The objects in a set are called the *elements*, or *members*, of the set.

A set is said to *contain* its elements.

Note:

1) Uppercase letters are usually used to denote sets, and lowercase letters are usually used to denote elements of sets.

2) $a \in A$: a is a member (an element) of the set A

$a \notin A$: a is not an element of the set A



The descriptions of a set

One can describe a set by

1) **Roster method: listing all its members between braces**

For example,

(1) The set of all odd positive integers less than 10 can be expressed by {1,3,5,7,9}

(2) $S = \{a, b, c, d\}$

2) **Brace notation with ellipses**

For example,

The set S of positive integers less than 100 can be denoted by $S = \{1, 2, 3, \ldots, 99\}$



3) Specification using set builder

$S = \{x \mid P(x)\}$

---- *S* contains all the elements from *U* which have the property *P*.

***Universal set* *U*:**

----The set contains all the objects under consideration.

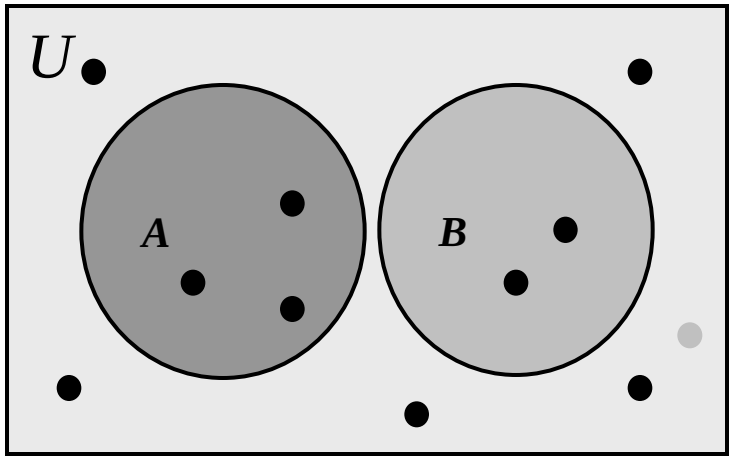
For example,

{*x* | *x* is an odd positive integer less than 10}



4) Venn diagrams

- **In Venn diagrams, the universal set *U* is represented by a rectangle.**
- **Inside this rectangle, circles or other geometrical figures are used to represent sets.**
- **Points are used to represent the particular elements of the set.**



Russell Paradox

Suppose there is a town with just one male barber; and that every man in the town keeps himself clean-shaven: some by shaving themselves, some by attending the barber. The barber obeys the following rule: he shaves all and only those men in town who do not shave themselves. Does the barber shave himself?



Note:

- **The Order of elements does not matter.**

  {1, 2, 3} = {3, 2, 1}

- **Repetition of elements does not matter.**

  {1, 1, 2, 3, 3} = {1, 2, 3}

〖Example 1〗Several sets and their notations are used throughout this book.

(1) R = {*x* | *x* is a real number}

(2) N = { *x* | *x* is a natural number}

(3) Z = { *x* | *x* is an integer}

(4) Z^+ = { *x* | *x* is a positive integer}

(5) ϕ -- The *void* set / *null* set / *empty* set



The relations between two sets

① Subset

$A \subseteq B$ ---- **A is a subset of the set B.**

$$A \subseteq B \Leftrightarrow \forall x\,(x \in A \rightarrow x \in B)$$

Problems:

For any set A,

(1) $\phi \subseteq A$?

Y.

The empty set is a subset of any set.

(2) $A \subseteq A$?

Y.



② Equal

$$A = B \Leftrightarrow \forall x[(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

$$\Leftrightarrow A \subseteq B \quad and \quad B \subseteq A$$

〖Example 2〗$S1 = \{a, b, c, d\}$, $S2 = \{a, d, b, c\}$, $S3 = \{b, c, a, d, d\}$.

$S1 = S2 = S3$?

③ Proper subset

$$A \subset B \Leftrightarrow \forall x\,(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

$$\Leftrightarrow A \subseteq B \quad and \quad A \neq B$$



【Definition】 Let *S* be a set. If there are exactly *n* distinct elements in *S* where *n* is a nonnegative integer, we say that *S* is a *finite* set and that *n* is the *cardinality* of *S*.

Notation:

|*S*|---- the cardinality of *S*

【Definition】A set is said to be *infinite* if it is not finite.

For example,

The set of positive integers is infinite.

〖Example 3〗

(1) Let *A* be the set of odd positive integers less than 10. Then | *A* | =?

(2) | ϕ | = ?



2. The power set

〖Example 4〗 Let $A=\{a,b,c\}$, find all subsets of the set *A*.

Solution:

ϕ

{*a*},{*b*},{*c*}

{*a*, *b*},{*b*, *c*},{*a*, *c*}

{*a*, *b*, *c*}

Problem: How many subsets of the set *A* are there?

$$C_3^0+C_3^1+C_3^2+C_3^3=8$$

In general, for a set *A* with *n* elements, the number of subsets is

$$C_n^0+C_n^1+C_n^2\cdots+C_n^n=\sum_{i=0}^{n}C_n^i=2^n$$



【Definition】Given a set *S*, the *power set* of *S* is the set of all subsets of the set *S*.

Notation:

***P*(*S*) ---- the power set of *S*.**

$$P(S) = \{x \mid x \subseteq S\}$$

Note:

(1) $|S|=n$ implies $|P(S)| = 2^n$

(2) *S* is finite and so is *P*(*S*).

(3)

$$\begin{cases} x \in P(S) \Rightarrow x \subseteq S \\ x \in S \Rightarrow \{x\} \in P(S) \\ S \in P(S) \end{cases}$$



〖Example 5〗 What is the power set of the set $\{\phi\} and \{\phi,\{\phi\}\}$ **?**

Solution:

(1) $S=\{\phi\}$

$$P(S)=\{\phi,\{\phi\}\},\qquad |P(S)|=2$$

(2) $S=\{\phi,\{\phi\}\}$

$$P(S)=\{\phi,\{\phi\},\{\{\phi\}\},\{\phi,\{\phi\}\}\},\qquad |P(S)|=4$$



〖Example 6〗 Show that $P(A) \in P(B) \Rightarrow A \in B$ **?**

Proof:

$$P(A) \in P(B) \Rightarrow P(A) \subseteq B$$

$$(A \in P(A)) \wedge (P(A) \subseteq B) \Rightarrow A \in B$$

Problem:

$$A \subseteq B \Rightarrow P(A) \subseteq P(B)?$$



3. Cartesian Products

【**Definition**】*The ordered n-tuple* $(a_1,a_2,\ldots,a_n)$ **is the ordered collection that has** a_1 **as its first element,** a_2 **as its second element, … , and** a_n **as its** n**th element.**

$(a_1, a_2, \ldots,a_n) =(b_1, b_2, \ldots, b_n) \Leftrightarrow a_i = b_i\ (i=1,2,\ldots, n)$

In particular, 2-tuples are called ***ordered pairs*****.**

Note:

If $x \neq y$, then $(x, y) \neq (y, x)$.

$(x, y) = (u, v) \Rightarrow x = u \quad and \quad y = v$



The *Cartesian product* of *A* and *B*: $A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$

The *Cartesian product* of $A_1, A_2, \ldots, A_n$:

$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i$ **for** $i=1, 2, \ldots, n\}$.



〖**Example 7**〗 $A=\{a,b\}, B=\{0,1,2\}$, $A\times B=?$ $B\times A=?$

solution:

$$A\times B=\{(a,0),(a,1),(a,2),(b,0),(b,1),(b,2)\}$$
$$B\times A=\{(0,a),(1,a),(2,a),(0,b),(1,b),(2,b)\}$$

Note:

- **If $|A|=m$, $|B|=n$, Then** $|A\times B|=|B\times A|=mn$
- $A\times B\neq B\times A$
- $A\times\phi=\phi\times A=\phi$
- $(x,y)\in A\times B\Rightarrow x\in A \ \wedge \ y\in B; (x,y)\notin A\times B\Rightarrow x\notin A \ \vee \ y\notin B$



5. Truth Sets of Quantifiers

Given a predicate P, and a domain D.

The *truth set* of P is the set of elements x in D for which $P(x)$ is true. namely,

$$\text{The } \textit{truth set} \text{ of } P = \{x \in D \mid P(x)\}$$



Homework (Due on March 21)

Ed. 8

Sec. 2.1 13, 20, 24, 26, 34(a),(c)

Ed. 7

Sec. 2.1 11, 18, 22, 24, 32(a),(c)



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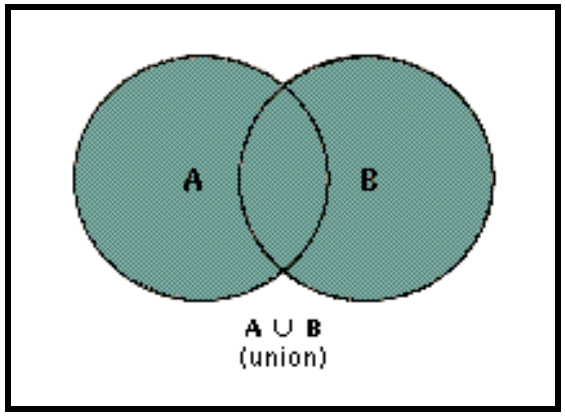
2.5 Cardinality of Sets



1. Set Operations

1) Union

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



According the Venn diagram,

$$(1) A \subseteq A \cup B, \qquad B \subseteq A \cup B$$

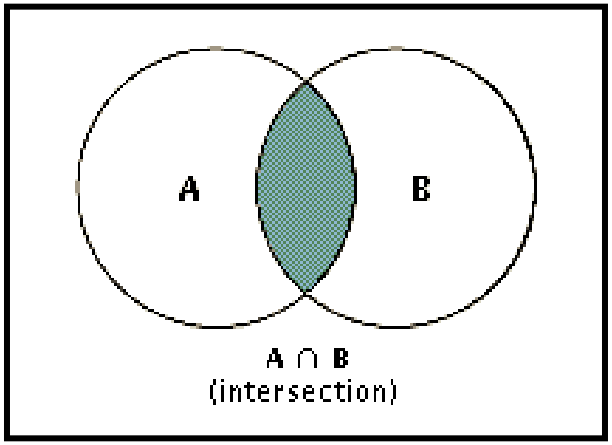
$$(2) A \subseteq C, B \subseteq C \Rightarrow A \cup B \subseteq C$$

$$(3) |A \cup B| \leq |A| + |B|$$

$$(4) A \cup B = B \Leftrightarrow A \subseteq B$$

2) Intersection

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



According the Venn diagram,

$$(1) A \cap B \subseteq A, \qquad A \cap B \subseteq B$$

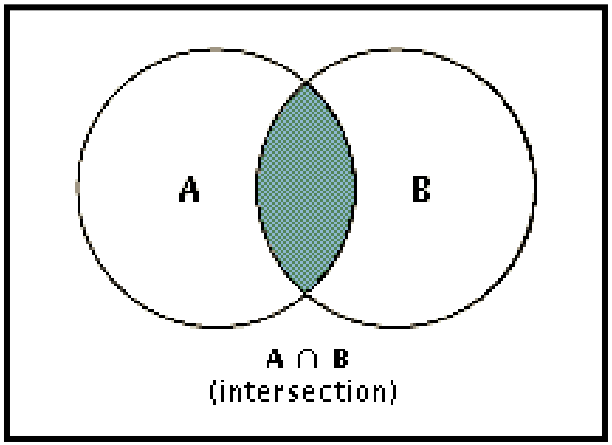
$$(2) C \subseteq A, C \subseteq B \Rightarrow C \subseteq A \cap B$$

$$(3) \mid A \cap B \mid \leq \mid A \mid, \mid A \cap B \mid \leq \mid B \mid$$

$$(4) A \cap B = A \Leftrightarrow A \subseteq B$$

2) Intersection

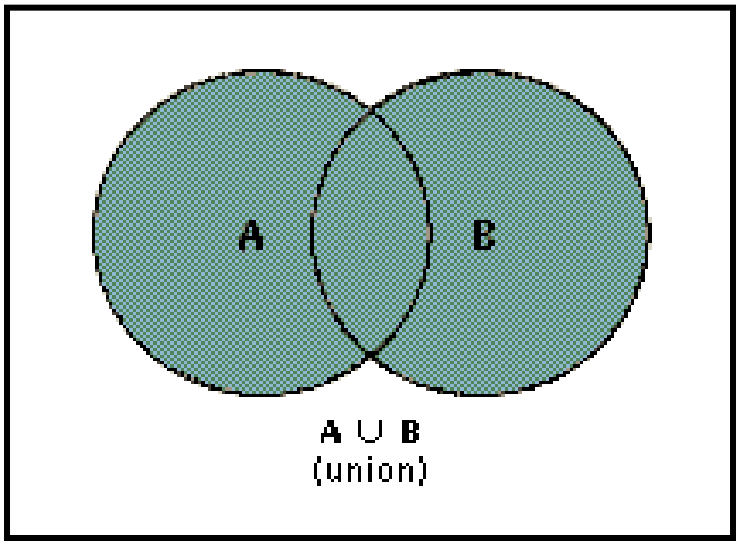
$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



Two sets are called *disjoint* if their intersection is the empty set, namely $A \cap B = \emptyset$

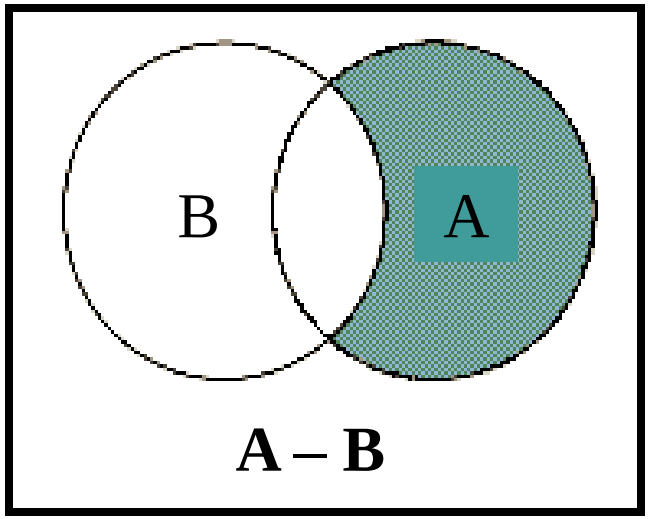
- **The cardinality of the union of two finite sets:**

$$|A \cup B| = |A| + |B| - |A \cap B|$$



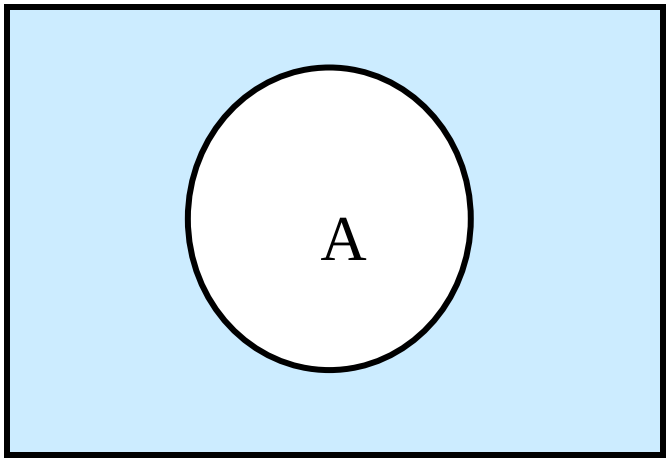
3) Difference of A and B

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$



4) the complement of a set

$$\overline{A} = \{x \mid x \notin A, x \in U\}$$



Where *U* is the universal set

Note: $A - B = A \cap \overline{B}$

5) Symmetric difference

$$A \oplus B = (A \cup B) - (A \cap B)$$

3. Set Identities

Set Identities	
Identity	Name
$A \cup \varnothing = A,\ A \cap U = A$	Identity laws
$A \cup U = U,\ A \cap \varnothing = \varnothing$	Domination laws
$A \cup A = A,\ A \cap A = A$	Idempotent laws
$\overline{\overline{A}} = A$	Complementation law
$A \cup B = B \cup A,\ A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws



Four ways to prove set identities

Show that $A \subseteq B$ and that $A \supseteq B$.

New & important

Use logical equivalences to prove equivalent set definitions.

Not hard, a little tedious

Use a membership table.

Like truth tables

Use previously proven identities.

Like $\equiv$



〖Example 1〗 Show that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ **?**

solution:

(1) $C = D$ iff $C \subseteq D$ and $D \subseteq C$

$\overline{(A \cup B)} \subseteq \overline{A} \cap \overline{B}$

Suppose that $x \in \overline{(A \cup B)}$ **.**

It follows that $x \notin A \cup B$ **.**

This implies that $x \notin A$ and $x \notin B$ **.**

Hence, $x \in \overline{A}$ and $x \in \overline{B}$ **.**

Thus $x \in \overline{A} \cap \overline{B}$ **.**

❖ $\overline{A} \cap \overline{B} \subseteq \overline{(A \cup B)}$ **(detail omitted)**

Q.E.D.



〖Example 1〗 Show that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ **?**

solution:

(2) Use set builder notation and logical equivalences

$$\begin{aligned}
\overline{(A \cup B)} &= \{x \mid x \notin A \cup B\} \\
&= \{x \mid \neg(x \in A \cup B)\} \\
&= \{x \mid \neg(x \in A \vee x \in B)\} \\
&= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\} \\
&= \{x \mid x \notin A \wedge x \notin B\} \\
&= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} \\
&= \{x \mid x \in \overline{A} \cap \overline{B}\} \\
&= \overline{A} \cap \overline{B}
\end{aligned}$$



〖Example 2〗 Let *A*, *B*, and *C* be sets. Show that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Membership Table

A	B	C	B∪C	A∩(B∪C)	A ∩ B	A ∩ C	(A ∩ B) ∪(A ∩ C)
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0



〖Example 3〗 Show that $(A-B)\cup B = A\cup B$ **?**

solution:

$$(A-B)\cup B = (A\cap \overline{B})\cup B$$

$$= (A\cup B)\cap(\overline{B}\cup B)$$

$$= (A\cup B)\cap U$$

$$= A\cup B$$



〖Example 4〗 Simplify the following set.

$$((A \cup B \cup C) \cap (A \cup B) - ((A \cup (B - C)) \cap A)$$

solution:

$$((A \cup B \cup C) \cap (A \cup B) - ((A \cup (B - C)) \cap A)$$

$$= (A \cup B) - A$$

$$= (A \cup B) \cap \overline{A}$$

$$= (A \cap \overline{A}) \cup (B \cap \overline{A})$$

$$= \phi \cup (B \cap \overline{A})$$

$$= B - A$$



〖**Example 5**〗Proof that if (A - B) U (B - A) = (A U B) then $\underline{A \cap B = \varnothing}$

a) $A \cup B = \varnothing$
b) $A = B$
c) $A \cap B = \varnothing$
d) $A - B = B - A = \varnothing$

Proof:

Suppose to the contrary, that $A \cap B \neq \varnothing$, and that $x \in A \cap B$.

Then *x* cannot be in A-B and *x* cannot be in B-A.

Then x is not in (A - B) U (B - A).

But x is in A U B since $(A \cap B) \subseteq (A \cup B)$.

Thus, $A \cap B = \varnothing$.



4. Generalized Unions and Intersections

$$A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^{n} A_i$$

$$A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^{n} A_i$$



5. Computer Representation of Set

Using bit strings to represent sets.

(1) Specify an arbitrary ordering of the elements of *U*,

for instance $a_1, a_2, \cdots, a_n$

(2) Represent a subset *A* of *U* with the bit string of length *n*, where the *i*th bit is 1 if a_i belongs to *A* and is 0 if a_i does not belong to *A*.

Union: bitwise OR

Intersection: bitwise AND



〖Example 5〗 Let U ={1, 2, 3, 4, 5,6,7,8,9}, A={1, 2, 3, 4, 5}, B= (1, 3, 5, 7, 9). Use bit strings to find the difference of A and B.

solution:

The bit string for the set A: 11 1110 000

The bit string for the set B: 10 1010 101

$$A - B = A \cap \overline{B}$$



Homework (Due on March 28)

Ed. 8

Sec. 2.2 19, 54, 63(c)

Ed. 7

Sec. 2.2 17, 48, 57(c)

